

1 Chern-Weil Homomorphism

Definition 1. Let G be a Lie group with its Lie algebra $\mathfrak{g}$, a polynomial $f : \mathfrak{g} \rightarrow \mathbb{K}$ is called an **Ad(G)-invariant polynomial** if for $\forall g \in G$ and $\forall X \in \mathfrak{g}$, we have

$$f(\text{Ad}(g)X) = f(X)$$

the set of all such polynomials will be denoted as $\text{Inv}(G)$. If $f \in \text{Inv}(G)$ is of degree k , we will denote it as $f \in \text{Inv}^k(G)$, hence

$$\text{Inv}(G) = \bigoplus_{k=0}^{\infty} \text{Inv}^k(G)$$

Remark: For a vector space V , by a **polynomial of degree k on V** , we mean an element $f \in \text{Sym}^k(V^*)$, here Sym means the symmetric tensor. So if $\{e_1, e_2, \dots, e_n\}$ is a basis of V and $\{e^1, e^2, \dots, e^n\}$ the dual basis of V^* , $f \in \text{Sym}^k(V^*)$ may be expressed as

$$f \triangleq \sum_{1 \leq i_1, i_2, \dots, i_k \leq n} a_{i_1 i_2, \dots, i_k} e^{i_1} e^{i_2} \dots e^{i_k}$$

Example 1. Let $G = GL_n(\mathbb{C})$, then we know that the adjoint representation is given by

$$\text{Ad}(g)X = gXg^{-1} \quad (\forall g \in GL_n(\mathbb{C}), \forall X \in \mathfrak{gl}_n(\mathbb{C}))$$

in which case $f \in \text{Inv}(GL_n(\mathbb{C}))$ is refer to

$$f(gXg^{-1}) = f(X)$$

1.1 On Principal Bundles

Recall: Recall that for a principal G -bundle $\pi : P \rightarrow M$, the connection on P is a $\mathfrak{g}$ -valued 1 form $\omega \in \Omega^1(P; \mathfrak{g})$ and the curvature $\Omega \in \Omega^2(P; \mathfrak{g})$ of the connection ω is given by $\Omega := d\omega + \frac{1}{2}[\omega, \omega]$ which satisfies some properties (refer [1] or [3]).

Let $\pi : P \rightarrow M$ be a principal G -bundle, let $f : \mathfrak{g} \rightarrow \mathbb{K}$ be a polynomial of degree k , i.e., $f \in \text{Sym}^k(\mathfrak{g}^*)$. Take a basis $\{e_1, \dots, e_n\}$ of $\mathfrak{g}$ and the dual basis $\{e^1, \dots, e^n\}$ of $\mathfrak{g}^*$, then f is given by

$$f \triangleq \sum_I a_I e^{i_1} e^{i_2} \dots e^{i_k}$$

where $I = (i_1, i_2, \dots, i_k)$ with $1 \leq i_1, i_2, \dots, i_k \leq n$. Now assume there is a given connection ω on P and hence the curvature Ω . So the curvature Ω is expressed as

$$\Omega \triangleq \sum_i \Omega^i \otimes e_i$$

then f induces a $2k$ -form $f(\Omega) \in \Omega^{2k}(P)$, which is defined as

$$f(\Omega) := \sum_I a_I \Omega^{i_1} \wedge \Omega^{i_2} \wedge \dots \wedge \Omega^{i_k}$$

Claim: The $2k$ -form $f(\Omega)$ defined above is independent of the choice of the basis $\{e_i\}_{i=1}^n$.

Proof: Let $\{e_i\}_{i=1}^n$ and $\{u_j\}_{j=1}^n$ be two basis of $\mathfrak{g}$ and the dual basis $\{\alpha^i\}_{i=1}^n, \{\beta^j\}_{j=1}^n$ of $\mathfrak{g}^*$ respectively. Assume that w.r.t. these two basis

$$f \triangleq \sum a_I \alpha^{i_1} \alpha^{i_2} \dots \alpha^{i_k}$$

$$f \triangleq \sum b_I \beta^{i_1} \beta^{i_2} \dots \beta^{i_k}$$

and assume that

$$\Omega = \sum_i \Omega^i \otimes e_i = \sum_j \Psi^j \otimes u_j$$

We just need to prove $\sum a_I \Omega^{i_1} \wedge \Omega^{i_2} \wedge \dots \wedge \Omega^{i_k} = \sum b_I \Psi^{i_1} \wedge \Psi^{i_2} \wedge \dots \wedge \Psi^{i_k}$:

Suppose

$$\beta^j = \sum a_i^j \alpha^i$$

note that $\Psi^j = \beta^j(\Omega)$ and $\omega^i = \alpha^i(\Omega)$, so we have

$$\Psi^j = \beta^j(\Omega) = \sum a_i^j \alpha^i(\Omega) = \sum a_i^j \Omega^i$$

therefore, we compute

$$\begin{aligned} f &= \sum b_I \beta^{i_1} \beta^{i_2} \dots \beta^{i_k} = \sum b_I (a_{j_1}^{i_1} \alpha^{j_1}) (a_{j_2}^{i_2} \alpha^{j_2}) \dots (a_{j_k}^{i_k} \alpha^{j_k}) \\ &= \sum b_I a_{j_1}^{i_1} a_{j_2}^{i_2} \dots a_{j_k}^{i_k} \alpha^{j_1} \alpha^{j_2} \dots \alpha^{j_k} \end{aligned}$$

so by the definition, we get

$$\begin{aligned} \sum a_I \Omega^{i_1} \wedge \Omega^{i_2} \wedge \dots \wedge \Omega^{i_k} &= f \left(\sum \Omega^i \otimes e_i \right) = \left(\sum b_I a_{j_1}^{i_1} a_{j_2}^{i_2} \dots a_{j_k}^{i_k} \alpha^{j_1} \alpha^{j_2} \dots \alpha^{j_k} \right) \left(\sum \Omega^i \otimes e_i \right) \\ &= \sum b_I a_{j_1}^{i_1} a_{j_2}^{i_2} \dots a_{j_k}^{i_k} \Omega^{j_1} \wedge \Omega^{j_2} \wedge \dots \wedge \Omega^{j_k} = \sum b_I (a_{j_1}^{i_1} \Omega^{j_1}) \wedge (a_{j_2}^{i_2} \Omega^{j_2}) \wedge \dots \wedge (a_{j_k}^{i_k} \Omega^{j_k}) \\ &= \sum b_I \Psi^{i_1} \wedge \Psi^{i_2} \wedge \dots \wedge \Psi^{i_k} \end{aligned}$$

which ensures the claim. $\square$

Recall: Recall that for a principal G -bundle $\pi : O \longrightarrow M$, we have the **covariant derivative** (see the appendix below or [1],[2]) D acting on $\Omega^k(P)$, which is given by

$$D\varphi(X_1, X_2, \dots, X_k) := d\varphi(h(X_1), h(X_2), \dots, h(X_k)) \quad (\forall \varphi \in \Omega^k(P), \forall X_i \in \Gamma(TP))$$

where $h : TP \longrightarrow \mathcal{H}$ is the projection to the horizontal distribution. And we simply denote this as $D\varphi = (d\varphi)^h$.

Lemma 1. Let $\pi : P \longrightarrow M$ be a principal G -bundle and a connection on P is given, let $\varphi \in \Omega^*(P)_{\text{bas}}$, i.e., φ is a basic form on P , then we have

$$D\varphi = d\varphi$$

i.e., the covariant derivative acting on the basic forms is the same as the exterior differential operator d , $D|_{\Omega^*(P)_{\text{bas}}} = d$.

Proof: Since giving a connection on P is equivalent to giving a horizontal distribution on P (see [1] or [3]), hence for any $X \in \Gamma(TP)$, it can be decomposed as

$$X = hX + vX$$

where $h : TP \rightarrow \mathcal{H}$, $v : TP \rightarrow \mathcal{V}$ are the projections respectively. Since the vertical tangent space at $p \in P$ is $\mathcal{V}_p := \ker(d\pi_p)$, we get

$$d\pi(X) = d\pi(hX)$$

i.e.,

$$d\pi = d\pi \circ h$$

Assume that $\varphi \in \Omega^k(P)_{\text{bas}}$, which means $\exists \alpha \in \Omega^k(M)$ such that $\varphi = \pi^*\alpha$ (see [4]). Thus

$$\begin{aligned} D\varphi &= d\varphi \circ h = (d\pi^*\alpha) \circ h = (\pi^*d\alpha) \circ h = d\alpha \circ d\pi \circ h = d\alpha \circ d\pi \\ &= \pi^*(d\alpha) = d(\pi^*\alpha) = d\varphi \end{aligned}$$

which proves the lemma. $\square$

Our main result on principal bundles is stated as below.

Theorem 1. Let $\pi : P \rightarrow M$ be a principal G -bundle with any given connection ω and hence its curvature Ω . Let $f \in \text{Inv}^k(G)$, then we have

- (1) $f(\Omega) \in \Omega^{2k}(P)_{\text{bas}}$, i.e., $\exists \Lambda \in \Omega^{2k}(M)$ such that $f(\Omega) = \pi^*\Lambda$;
- (2) $\Lambda \in \ker(d|_{\Omega^{2k}(M)})$;
- (3) $[\Lambda] \in H_{\text{dR}}^{2k}(M)$ is independent of the choice of the connection ω on P .

Proof: (1) We need to check that $f(\Omega)$ is both G -invariant and horizontal (see [1] or [4]). For any $g \in G$, by the definition of $f(\Omega)$, we have

$$r_g^*f(\Omega) = r_g^*\left(\sum a_I \Omega^{i_1} \wedge \cdots \wedge \Omega^{i_k}\right) = \sum (a_I r_g^* \Omega^{i_1} \wedge \cdots \wedge r_g^* \Omega^{i_k})$$

here note that $r_g^*a_I = a_I$ since a_I are constant coefficient. By the property of the curvature $\Omega \in \Omega^2(P; \mathfrak{g})$, we know that

$$r_g^*\Omega = (\text{Ad}g^{-1}) \circ \Omega$$

i.e., if $\Omega \triangleq \sum \Omega^i \otimes e_i$, then

$$r_g^*\left(\sum \Omega^i \otimes e_i\right) = \sum ((\text{Ad}g^{-1}) \circ \Omega)^i \otimes e_i$$

hence we have

$$r_g^*\Omega^i = ((\text{Ad}g^{-1}) \circ \Omega)^i$$

so we compute

$$\begin{aligned} r_g^*f(\Omega) &= \sum a_I (r_g^* \Omega^{i_1}) \wedge \cdots \wedge (r_g^* \Omega^{i_k}) \\ &= \sum a_I (\text{Ad}g^{-1} \circ \Omega)^{i_1} \wedge \cdots \wedge (\text{Ad}g^{-1} \circ \Omega)^{i_k} = f(\text{Ad}g^{-1} \circ \Omega) \end{aligned}$$

since $f \in \text{Inv}^k(G)$, it's clear to see that

$$f(\text{Ad}g^{-1} \circ \Omega) = f(\Omega)$$

which means

$$r_g^* f(\Omega) = f(\Omega)$$

i.e., f is G -invariant. By the property of Ω , we know that $\Omega(X, Y) = (d\omega)(h(X), h(Y))$ (see [1] or [3]) for all $X, Y \in \Gamma(TP)$, which implies that

$$\Omega \in \Omega^2(P; \mathfrak{g})_{\text{hor}}$$

i.e., Ω is horizontal (see [1] or [3]), so we immediately get

$$\Omega^i \in \Omega^2(P)_{\text{hor}}$$

i.e.,

$$i(V)\Omega^i = 0 \quad (\forall V \in \Gamma(\mathcal{V}))$$

then we get

$$i(V)f(\Omega) = i(V) \left(\sum a_I \Omega^{i_1} \wedge \cdots \wedge \Omega^{i_k} \right) = 0 \quad (\forall V \in \Gamma(\mathcal{V}))$$

so $f(\Omega)$ is also horizontal. Since $f(\Omega)$ is both G -invariant and horizontal, then it is basic (see [4]), i.e., $\exists \Lambda \in \Omega^{2k}(M)$ such that $f(\Omega) = \pi^* \Lambda$;

(2) Note that $\pi : P \rightarrow M$ is surjective and hence so is the differential map $d\pi : TP \rightarrow TM$. Also, we see that the pullback map $\pi^* : \Omega^*(M) \rightarrow \Omega^*(P)$ is injective by its contravariant functoriality. So if we can prove that $\pi^*(d\Lambda) = 0$ then we immediately get $d\Lambda = 0$ from the injectivity of π^* . Now we prove this:

note that

$$\pi^*(d\Lambda) = d(\pi^* \Lambda) = d(f(\Omega))$$

we express $f(\Omega)$ as

$$f(\Omega) \triangleq \sum_I a_I \Omega^{i_1} \wedge \cdots \wedge \Omega^{i_k}$$

Since we know that $D\Omega = 0$ (see the appendix below or [1] or [2]), this means

$$D\Omega^i = 0 \quad (\forall i)$$

note that by the property of the curvature $\Omega \in \Omega^2(P; \mathfrak{g})$ (see [1] or [3] and the appendix below)

$$\Omega \in \Omega_{\text{Ad}}^2(P; \mathfrak{g})$$

this means that if $\Omega \triangleq \Omega^i \otimes e_i$, then we have

$$\Omega^i \in \Omega_{\text{Ad}}^2(P) \quad (\forall i)$$

by (1) we know that $f(\Omega)$ is basic, so according to Lemma 1, we compute

$$d(f(\Omega)) = D(f(\Omega)) = D \left(\sum_I a_I \Omega^{i_1} \wedge \cdots \wedge \Omega^{i_k} \right)$$

here note that the coefficients a_I are all constants, hence

$$a_I \in \Omega^0(P)_{\text{bas}}$$

so by Lemma 1, we have

$$Da_I = da_I = 0$$

because D is an antiderivation of degree +1 on the tensorial forms (see the appendix below)

$$d(f(\Omega)) = D \left(\sum_I a_I \Omega^{i_1} \wedge \cdots \wedge \Omega^{i_k} \right) = \sum_I \sum_j a_I \Omega^{i_1} \wedge \cdots \wedge D\Omega^{i_j} \wedge \cdots \wedge \Omega^{i_k} = 0$$

since each $D\Omega^{i_j} = 0$. So we see that

$$d(f(\Omega)) = \pi^*(d\Lambda) = 0$$

so by the injectivity of $\pi^* : \Omega^*(M) \longrightarrow \Omega^*(P)$, $d\Lambda=0$;

(3) Given any two connections $\omega_0, \omega_1 \in \Omega^1(P; \mathfrak{g})$ on P , for any $t \in [0, 1]$, define a $\mathfrak{g}$ -valued 1-form on $P \times I$ as follows

$$\widehat{\omega} := (1-t)\pi_1^*\omega_0 + t\pi_1^*\omega_1 \quad (1)$$

where $\pi_1 : P \times I \longrightarrow P$ is the projection to the first factor and $I(\subseteq \mathbb{R})$ is an open interval containing $[0, 1]$. We first check that for any $t \in [0, 1]$, $\widehat{\omega}$ is a connection on the principal bundle $P \times I \longrightarrow M \times I$:

Take $\forall A \in \mathfrak{g}$, note that $\widetilde{\pi} : P \times I \longrightarrow M \times I$ is a principal G -bundle, where the action is given by $(p, t)g = (pg, t)$. We denote the fundamental vector field induced by $A \in \mathfrak{g}$ on $P \times I$ by $\underline{A}^{P \times I} \in \Gamma(T(P \times I))$, so for any point $(p, t) \in P \times I$, we have

$$(\pi_1^*\omega_0)_{(p,t)} \left(\underline{A}_{(p,t)}^{P \times I} \right) = (\omega_0)_p \left((d\pi_1)_{(p,t)}(\underline{A}_{(p,t)}^{P \times I}) \right)$$

so by the definition of the fundamental vector field, we have

$$\begin{aligned} (d\pi_1)_{(p,t)}(\underline{A}_{(p,t)}^{P \times I}) &= (d\pi_1)_{(p,t)} \left(\frac{d}{ds} \Big|_{s=0} (p, t) \exp(sA) \right) \\ &= \frac{d}{ds} \Big|_{s=0} \left[\pi_1 \left((p, t) \exp(sA) \right) \right] \end{aligned}$$

note that since π_1 is the projection, so it is a G -equivariant map, i.e., $(\pi_1(p, t))g = \pi_1((p, t)g)$, so back to our computation, it gives

$$(d\pi_1)_{(p,t)}(\underline{A}_{(p,t)}^{P \times I}) = \frac{d}{ds} \Big|_{s=0} \left[\left(\pi_1(p, t) \right) \exp(sA) \right] = \frac{d}{ds} \Big|_{s=0} (p \exp(sA)) = \underline{A}_p^P$$

hence we get

$$(\pi_1^*\omega_0)_{(p,t)} \left(\underline{A}_{(p,t)}^{P \times I} \right) = (\omega_0)_p(\underline{A}_p^P) = A$$

a same discussion shows that

$$(\pi_1^*\omega_1)_{(p,t)} \left(\underline{A}_{(p,t)}^{P \times I} \right) = A$$

so for any point $(p, t) \in P \times I$, any $A \in \mathfrak{g}$ and any $t' \in [0, 1]$, we have

$$\widehat{\omega}_{(p,t)} \left(\underline{A}_{(p,t)}^{P \times I} \right) = \left((1-t')\pi_1^*\omega_0 + t'\pi_1^*\omega_1 \right)_{(p,t)} \left(\underline{A}_{(p,t)}^{P \times I} \right) = (1-t')A + t'A = A$$

finally, just recall that what we have done in the proof of the the properties that $r_g^*\omega = \text{Ad}(g^{-1}) \circ \omega$ for the connection $\omega \in \Omega^1(P; \mathfrak{g})$ (see [1] or [3]:thm2), if we replace ω as $\pi_1^*\omega_0$ and $\pi_1^*\omega_1$ in our case, then the same discussion also leads to

$$r_g^*(\pi_1^*\omega_0) = \text{Ad}(g^{-1}) \circ (\pi_1^*\omega_0), \quad r_g^*(\pi_1^*\omega_1) = \text{Ad}(g^{-1}) \circ (\pi_1^*\omega_1)$$

where we just need to note that the right-translation r_g on $P \times I$ is given by $(p, t) \mapsto (pg, t)$. Hence for all $t' \in [0, 1]$ and all $g \in G$ we get

$$\begin{aligned} r_g^*\widehat{\omega} &= (1 - t')r_g^*(\pi_1^*\omega_0) + t'r_g^*(\pi_1^*\omega_1) = (1 - t')\text{Ad}(g^{-1}) \circ (\pi_1^*\omega_0) + t'\text{Ad}(g^{-1}) \circ (\pi_1^*\omega_1) \\ &= \text{Ad}(g^{-1}) \circ ((1 - t')(\pi_1^*\omega_0) + t'(\pi_1^*\omega_1)) = \text{Ad}(g^{-1}) \circ \widehat{\omega} \end{aligned}$$

So in conclusion, the form defined by (1) is indeed a connection on the princial bundle $\widetilde{\pi} : P \times I \longrightarrow M \times I$ for $\forall t \in [0, 1]$.

Now, for every $t \in [0, 1]$ we define the inclusion operator

$$i_t : P \hookrightarrow P \times I$$

$$p \mapsto (p, t)$$

note that for $\forall t \in [0, 1]$, we have $\pi_1 \circ i_t = \text{Id}$, hence we immediately get

$$(\pi \circ i_t)^* = i_t^* \circ \pi_1^* = \text{Id} \quad (\forall t \in [0, 1])$$

which implies that

$$i_t^*\widehat{\omega} = (1 - t)i_t^*\pi_1^*\omega_0 + ti_t^*\pi_1^*\omega_1 = (1 - t)\omega_0 + t\omega_1$$

thus

$$i_0^*\widehat{\omega} = \omega_0, \quad i_1^*\widehat{\omega} = \omega_1$$

Now, let

$$\widehat{\Omega} = d\widehat{\omega} + \frac{1}{2}[\widehat{\omega}, \widehat{\omega}]$$

be the curvature of $\widehat{\omega}$ for every $t \in [0, 1]$. So we compute

$$i_t^*\widehat{\Omega} = i_t^*d\widehat{\omega} + i_t^*\left(\frac{1}{2}[\widehat{\omega}, \widehat{\omega}]\right) = d(i_t^*\widehat{\omega}) + \frac{1}{2}[i_t^*\widehat{\omega}, i_t^*\widehat{\omega}]$$

this means

$$i_0^*\widehat{\Omega} = d\omega_0 + \frac{1}{2}[\omega_0, \omega_0] = \Omega_0, \quad i_1^*\widehat{\Omega} = d\omega_1 + \frac{1}{2}[\omega_1, \omega_1] = \Omega_1 \quad (2)$$

Now, for $f \in \text{Inv}^k(G)$, if we express f as

$$f \triangleq \sum_I a_I e^{i_1} e^{i_2} \dots e^{i_k}$$

then we can see that

$$i_t^*f(\widehat{\Omega}) = i_t^*\left(\sum_I a_I \widehat{\Omega}^{i_1} \wedge \dots \wedge \widehat{\Omega}^{i_k}\right)$$

note that a_I are constant coefficients, we get

$$i_t^* f(\widehat{\Omega}) = \sum_I a_I (i_t^* \widehat{\Omega}^{i_1}) \wedge \cdots \wedge (i_t^* \widehat{\Omega}^{i_k})$$

so take $t = 0$ and $t = 1$ respectively, by equation (2), we get

$$i_0^* f(\widehat{\Omega}) = \sum_I a_I \Omega_0^{i_1} \wedge \cdots \wedge \Omega_0^{i_k} = f(\Omega_0)$$

and

$$i_1^* f(\widehat{\Omega}) = \sum_I a_I \Omega_1^{i_1} \wedge \cdots \wedge \Omega_1^{i_k} = f(\Omega_1)$$

By the definition of the operators i_t , we know that

$$i_0 \simeq i_1 : P \hookrightarrow P \times I$$

i.e., i_0 and i_1 are homotopic. Since homotopic maps induce the same homomorphism on cohomological groups (see [6]), we get

$$\begin{aligned} i_0^* &= i_1^* : H_{\text{dR}}^{2k}(P \times I) \longrightarrow H_{\text{dR}}^{2k}(P) \\ [f(\widehat{\Omega})] &\mapsto [i_0^* f(\widehat{\Omega})] = [i_1^* f(\widehat{\Omega})] \end{aligned}$$

which is saying

$$[f(\Omega_0)] = [f(\Omega_1)]$$

i.e.,

$$\pi^*[\Lambda_1] = [\pi^* \Lambda_1] = [\pi^* \Lambda_2] = \pi^*[\Lambda_2]$$

just as what we have said before, the pullback map

$$\pi^* : \Omega^*(M) \longrightarrow \Omega^*(P)$$

is injective, which implies that

$$[\Lambda_1] = [\Lambda_2]$$

i.e., independent of the choice of the connection ω on P . $\square$

Due to this theorem, we have the following definition.

Definition 2. Let $\pi : P \longrightarrow M$ be a principal G -bundle with any given connection ω and hence its curvature Ω . For any

$$f \triangleq \sum_k f_{[k]} \in \text{Inv}(G) = \bigoplus_{k=0}^{\infty} \text{Inv}^k(G)$$

there exists $\Lambda \triangleq \sum_k \Lambda_{[2k]} \in \Omega^{\text{even}}(M) := \bigoplus_k \Omega^{2k}(M)$ such that

$$f_{[k]}(\Omega) = \pi^* \Lambda_{[2k]} \in \Omega^{2k}(P), \quad f(\Omega) = \pi^* \Lambda = \sum_k \pi^* \Lambda_{[2k]} \in \Omega^{\text{even}}(P)$$

the cohomology class

$$[\Lambda] = \sum_k [\Lambda_{[2k]}] \in H_{\text{dR}}^{\text{even}}(M)$$

is called the **characteristic class of P associated to f** .

Remark: For any given connection ω on the principal bundle $\pi : P \longrightarrow M$, if we define the map

$$c_P : \text{Inv}(G) \longrightarrow H_{\text{dR}}^{\text{even}}(M)$$

$$f \mapsto [\Lambda]$$

where $[\Lambda]$ is described in the definition above, then it's clear to see that this map is an algebra homomorphism, called **Chern-Weil homomorphism**, i.e., $c_P(fg) = [\Lambda^f \wedge \Lambda^g] = [\Lambda^f] \wedge [\Lambda^g]$.

2 Appendix A

This appendix is made just in order to make the note more self-contained. Stated without proof. ~~because I will feel so bored to type so many codes.~~ (you may also see these materials in [1] or [2], where they are discussed very clear)

2.1 Associated Bundle

Notaion: For the representation $\rho : G \longrightarrow GL(V)$ of a group G , where V is a vector space, we will simply denote $gv \triangleq g \cdot v := \rho(g)(v)$ for any $g \in G$ and any $v \in V$.

Let $\pi : P \longrightarrow M$ be a principal G -bundle. Given any representation of G , $\rho : G \longrightarrow \text{Aut}(V)$ on a finite-dimensional vector space V , it induces the **associated bundle** $P \times_{\rho} V := P \times V / \sim$, where the equivalence relation $\sim$ is defined by

$$(p, v) \sim (p', v') \iff \exists g \in G \text{ such that } p' = pg, v' = g^{-1}v$$

Theorem 2.1. : The associated bundle $\hat{\pi} : P \times_{\rho} V \longrightarrow M$, $[p, v] \mapsto \pi(p)$ is a well-defined fiber bundle with fiber V on the base manifold M . Hence it can be viewed as a vector bundle if V is of finite-dimension.

2.2 Tensorial Forms

Let $\pi : P \longrightarrow M$ be a principal G -bundle and $\rho : G \longrightarrow \text{Aut}(V)$ be a finite-dimensional representation of G .

Definition 2.1. : $\alpha \in \Omega^k(P; V)$ is called **right-equivariant of type ρ** if for any $g \in G$ we have

$$r_g^* \alpha = \rho(g^{-1}) \circ \alpha$$

where $r_g : P \longrightarrow P$, $p \mapsto pg$ is the right translation.

Definition 2.2. : $\alpha \in \Omega^k(P; V)$ is called **horizontal** if for any $X \in \Gamma(\mathcal{V})$ we have

$$i(X)\alpha = 0$$

in which case we will denote as $\alpha \in \Omega^k(P; V)_{\text{hor}}$. (also see [4])

Definition 2.3. : $\alpha \in \Omega^k(P; V)$ is called **tensorial of type ρ** if for any $g \in G$ and any $X \in \Gamma(\mathcal{V})$ we have

$$r_g^* \alpha = \rho(g^{-1}) \circ \alpha \text{ and } i(X)\alpha = 0$$

i.e., α satisfies both of the conditions in the two definitions above. In this case, we denote as $\alpha \in \Omega_\rho^k(P; V)$.

2.3 Covariant Derivative

Definition 2.4. : For any $\alpha \in \Omega^k(P; V)$, the **covariant derivative D** acting on α is defined by

$$(D\alpha)(X_1, X_2, \dots, X_k) := d\alpha(h(X_1), h(X_2), \dots, h(X_k))$$

for any $X_i \in \Gamma(TP)$.

The important fact is the following theorem.

Theorem 2.2. : The covariant derivative D restricting on $\Omega_\rho^k(P; V)$ gives us an antiderivation of degree +1

$$D : \Omega_\rho^k(P; V) \longrightarrow \Omega_\rho^{k+1}(P; V)$$

i.e., for any $\omega, \tau \in \Omega_\rho^*(P; V)$, we have

$$D(\omega \wedge \tau) = (D\omega) \wedge \tau + (-1)^{|\omega|} \omega \wedge (D\tau)$$

where we denote $|\omega| := \deg \omega$. Here, we need to note that $D \circ D \neq 0$ in general, which is quite different from d .

Recall: Recall that for a Lie algebra $\mathfrak{g}$ together with its Lie bracket $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \longrightarrow \mathfrak{g}$, we can define the multiplication operation as follows

$$\Omega^k(P; \mathfrak{g}) \times \Omega^l(P; \mathfrak{g}) \longrightarrow \Omega^{k+l}(P; \mathfrak{g})$$

$$(\omega, \tau) \mapsto [\omega, \tau]$$

where for any $X_i \in \Gamma(TP)$, we set

$$[\omega, \tau](X_1, \dots, X_{k+l}) := \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} (\text{sgn} \sigma) [\omega(X_{\sigma(1)}, \dots, X_{\sigma(k)}), \tau(X_{\sigma(k+1)}, \dots, X_{\sigma(k+l)})]$$

Now, if we are given a Lie group representation $\rho : G \longrightarrow \text{Aut}(V)$, then we can also define the following multiplication operation

$$\Omega^k(P; \mathfrak{g}) \times \Omega^l(P; V) \longrightarrow \Omega^{k+l}(P; V)$$

$$(\omega, \tau) \mapsto \omega \cdot \tau$$

where for any $X_i \in \Gamma(TP)$, we set

$$(\omega \cdot \tau)(X_1, \dots, X_{k+l}) := \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} (\text{sgn} \sigma) \rho_*(\omega(X_{\sigma(1)}, \dots, X_{\sigma(k)})) (\tau(X_{\sigma(k+1)}, \dots, X_{\sigma(k+l)}))$$

where $\rho_* : \mathfrak{g} \longrightarrow \mathfrak{gl}(V)$ is the Lie algebra homomorphism induced by ρ . Thus for the principal G -bundle $\pi : P \longrightarrow M$, if we consider the case that $V := \mathfrak{g}$ and $\rho := \text{Ad} : G \longrightarrow \text{Aut}(\mathfrak{g})$, since we know that

$$\text{Ad}_* = \text{ad} : \mathfrak{g} \longrightarrow \mathfrak{gl}(\mathfrak{g})$$

where $\text{ad}(X)(Y) = [X, Y]$ for all $X, Y \in \mathfrak{g}$, so by our definition, we immediately get

$$(\omega \cdot \tau)(X_1, \dots, X_{k+l}) = \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} (\text{sgn} \sigma) [\omega(X_{\sigma(1)}, \dots, X_{\sigma(k)}), \tau(X_{\sigma(k+1)}, \dots, X_{\sigma(k+l)})]$$

compared with the multiplication on $\Omega^*(P; \mathfrak{g})$, it is also saying that

$$\omega \cdot \tau = [\omega, \tau]$$

in this case.

The covariant derivative D acting on the tensorial forms $\Omega_\rho^*(P; V)$ can be given as what the following theorem says.

Theorem 2.3. : For the principal G -bundle $\pi : P \longrightarrow M$ with a given connection ω and let $\rho : G \longrightarrow \text{Aut}(V)$ be a finite-dimensional representation of G . Then for $\forall \alpha \in \Omega_\rho^k(P; V)$, we have

$$D\alpha = d\alpha + \omega \cdot \alpha$$

i.e., $D|_{\Omega_\rho^*(P; V)} = d + \omega \cdot$

Corollary 2.1. : If the representation is $\rho = \text{Ad} : G \longrightarrow \text{Aut}(\mathfrak{g})$, then we have

$$D|_{\Omega_{\text{Ad}}^*(P; \mathfrak{g})} = d + [\omega, \cdot]$$

Note that for any given connection ω on the principal bundle, the curvature defined by $\Omega := d\omega + \frac{1}{2}[\omega, \omega]$ satisfies $\Omega(X, Y) = (d\omega)(h(X), h(Y))$ for $\forall X, Y \in \Gamma(TP)$, which means that

$$\Omega = D\omega$$

hence we get

$$D\omega = d\omega + \frac{1}{2}[\omega, \omega]$$

which implies that the connection $\omega \notin \Omega_{\text{Ad}}^1(P; \mathfrak{g})$ by the corollary above. Also note that $\Omega \in \Omega^2(P; \mathfrak{g})_{\text{hor}}$ thus from the second Bianchi identity, $d\Omega = [\omega, \Omega]$, immediately by the corollary above we get

$$D\Omega = d\Omega + [\omega, \Omega] = 0$$

which is an equivalent form of second Bianchi identity.

3 References

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